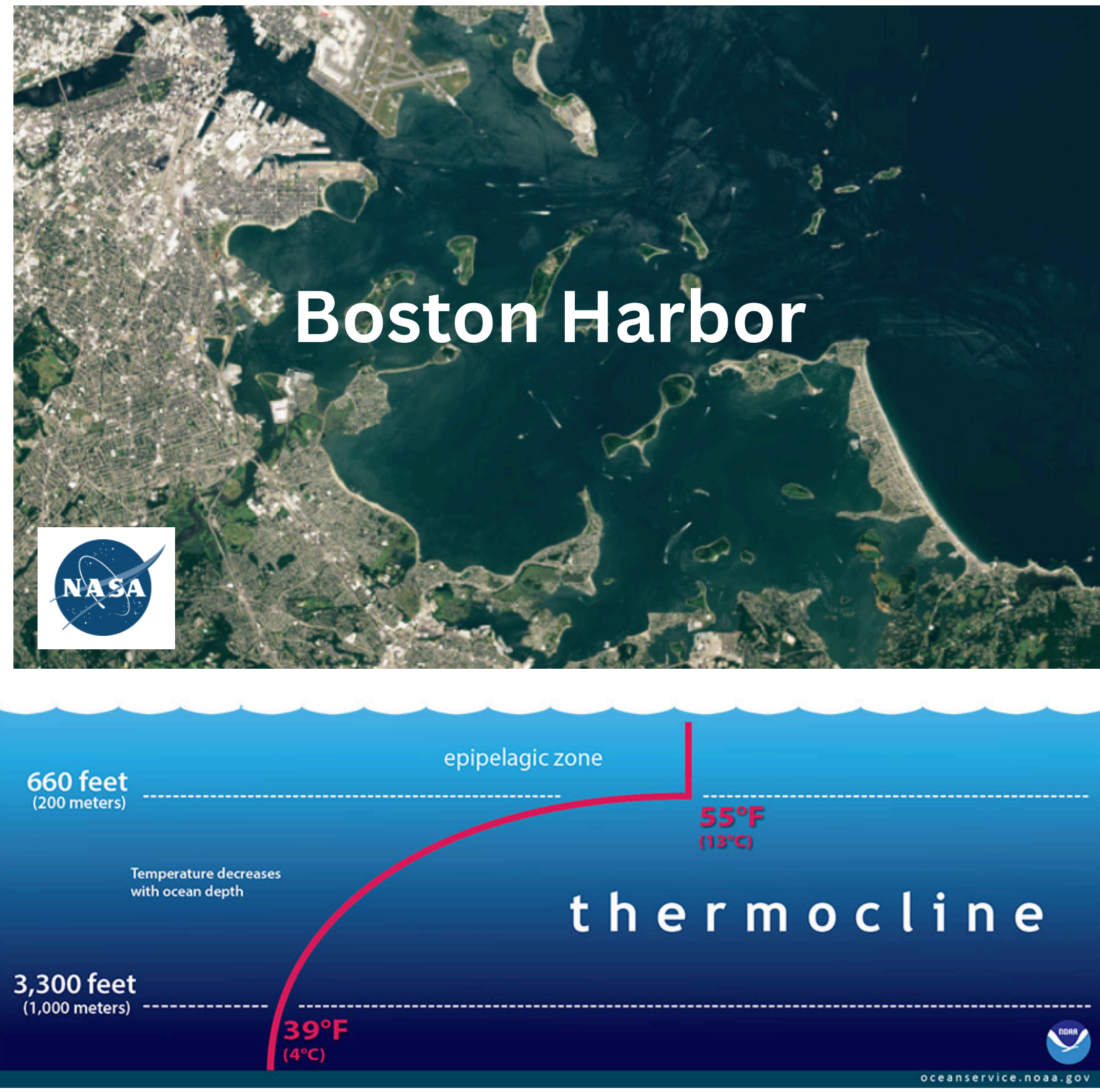




Improving temperature forecasts in Boston Harbor by combining satellite and sensor data

MOTIVATION

- The Boston Harbor (USA) is going through ecological and industrial change driven by urban development and climate change
- Sea Surface Temperature (SST) strongly affects marine ecosystems, water quality, and harbor industries
- Gaussian Process Regression (GPR) combines different data sources to improve prediction and to help understand the uncertainty in SST



Thermoclines

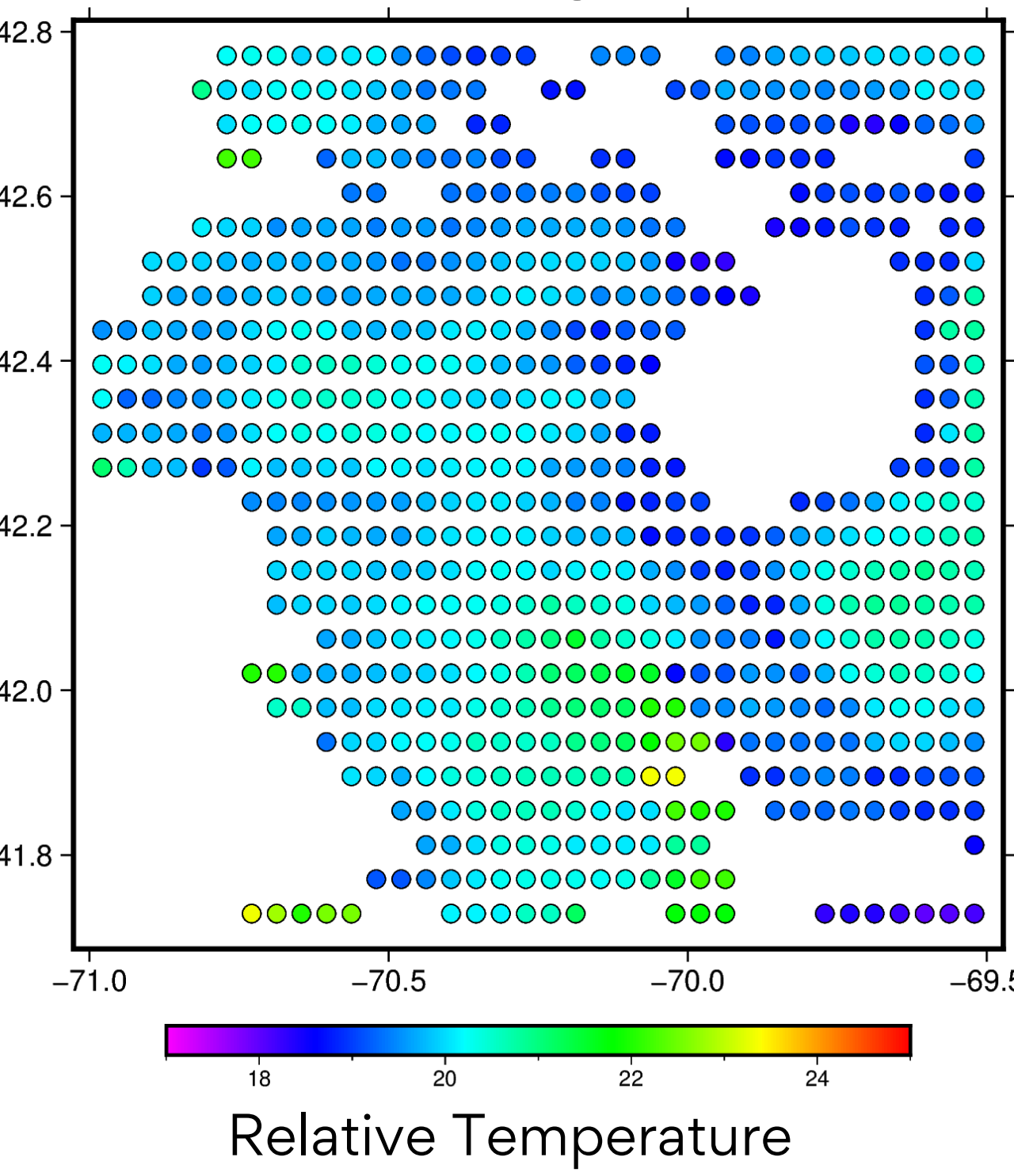
- Thermoclines create subsurface temperature gradients
- The available data to us only samples the surface, not accounting for thermocline

OBJECTIVES

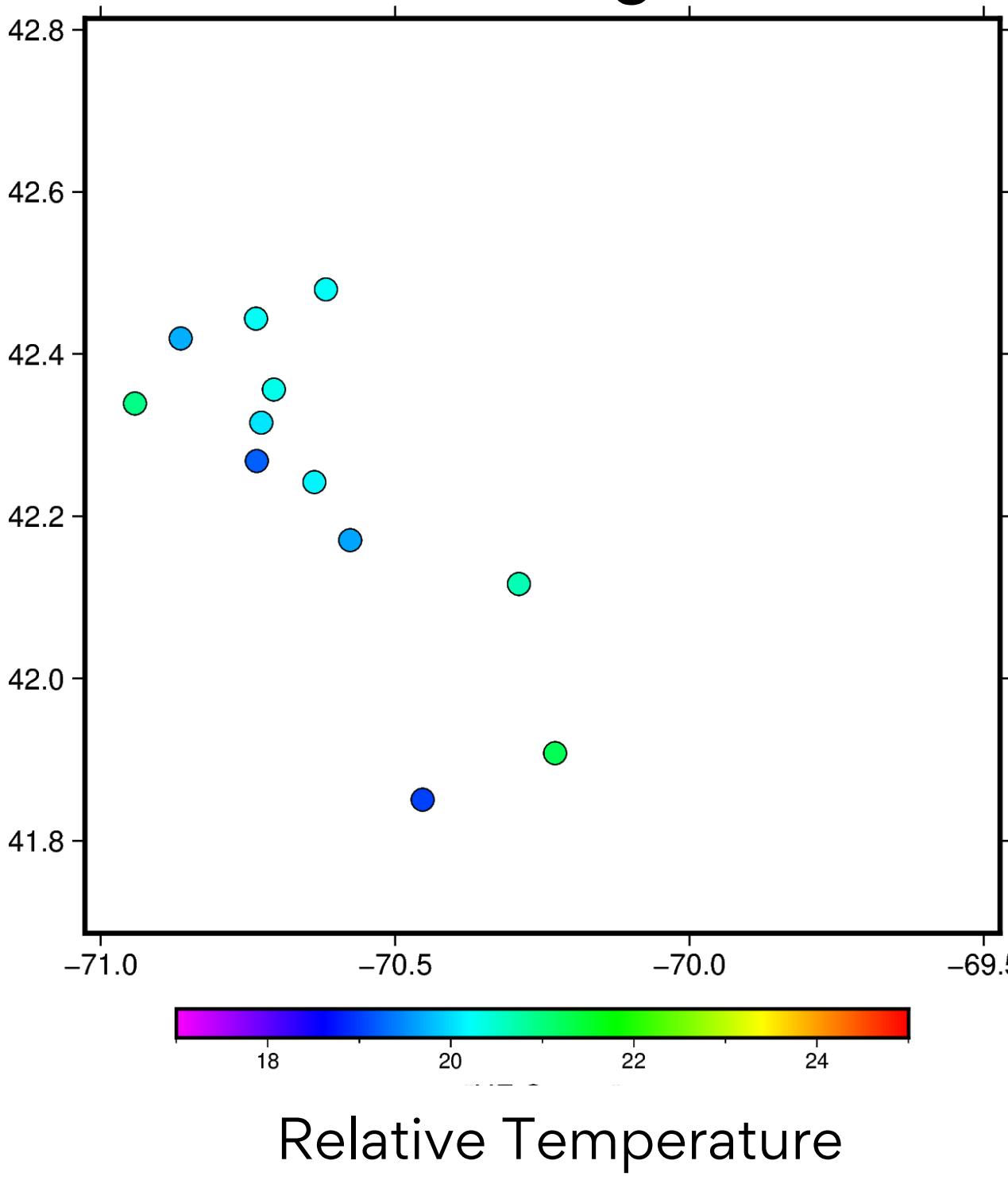
- Develop a multi-fidelity GPR model (Meng et al., 2021) that uses both low-fidelity (LF) satellite and high-fidelity (HF) buoy sensor temperature data
- Evaluate the model's accuracy in predicting temperatures across the Boston Harbor
- Compare model performance against single-fidelity GPR

DATA

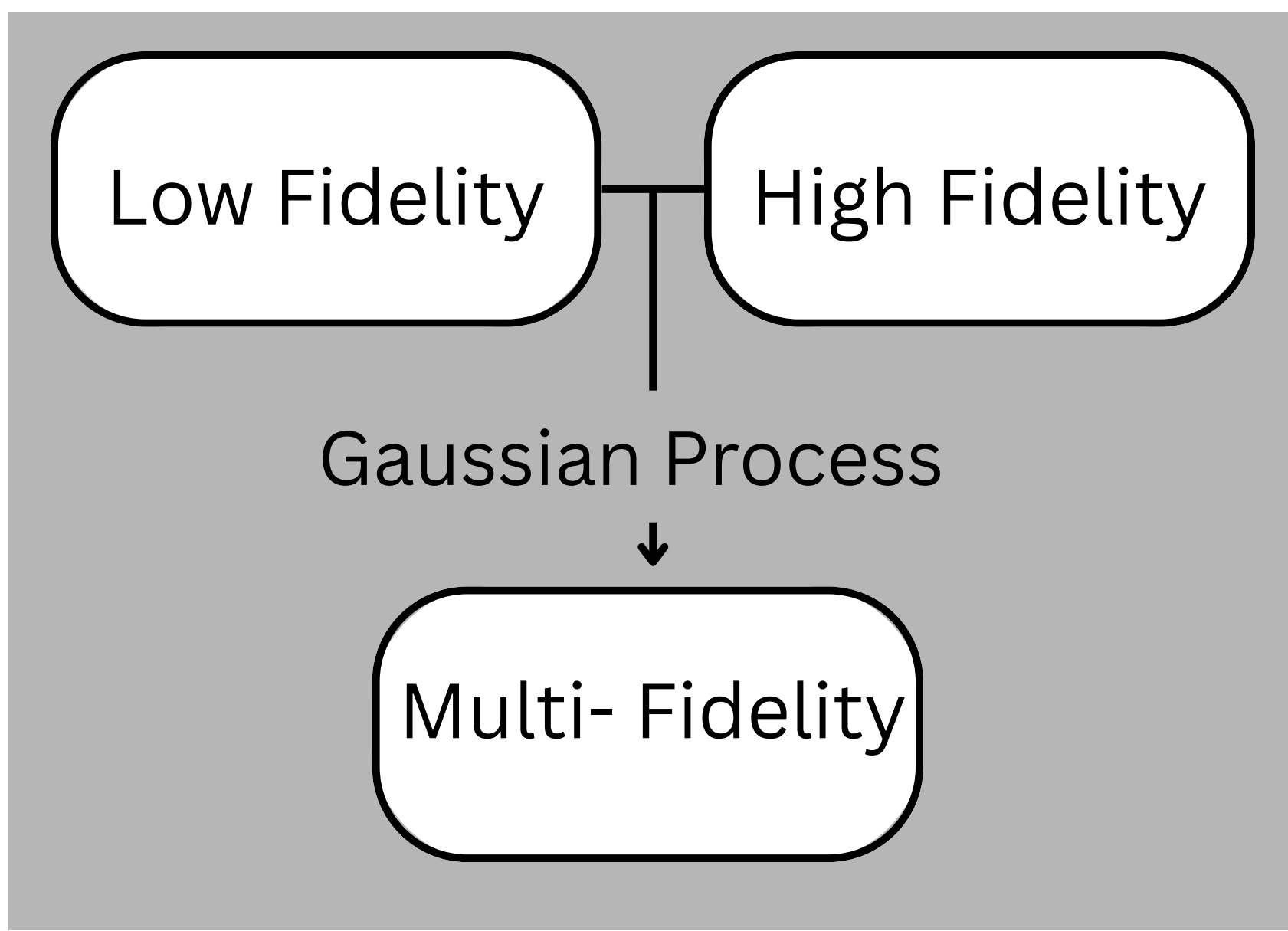
LF Training Points



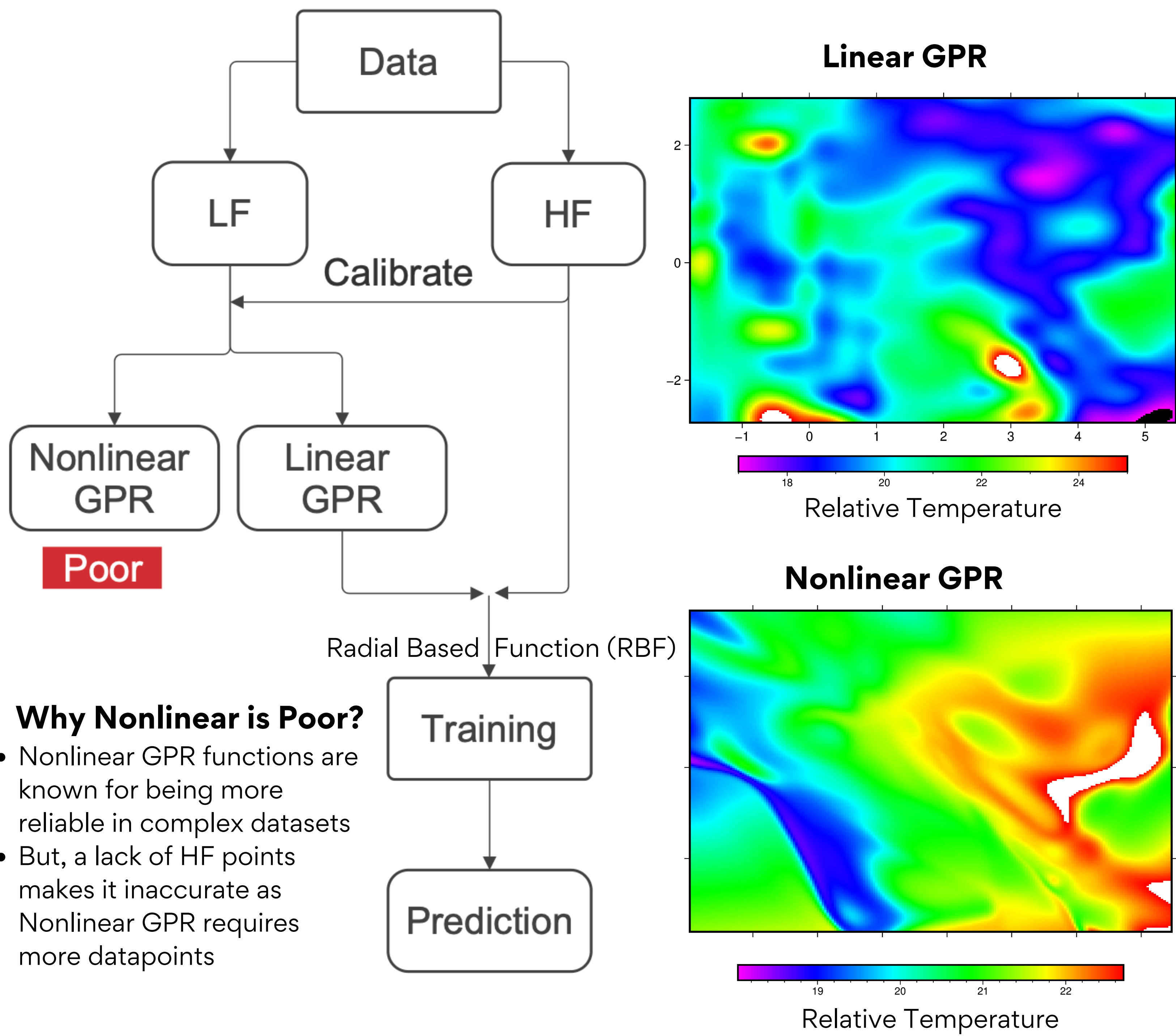
HF Training Points



Property	Low Fidelity	High Fidelity
Source	Satellite	Buoy Sensor
Total Data	653	12
Validation Data	12	2
Reliability	Low	High
Cost	Low	High

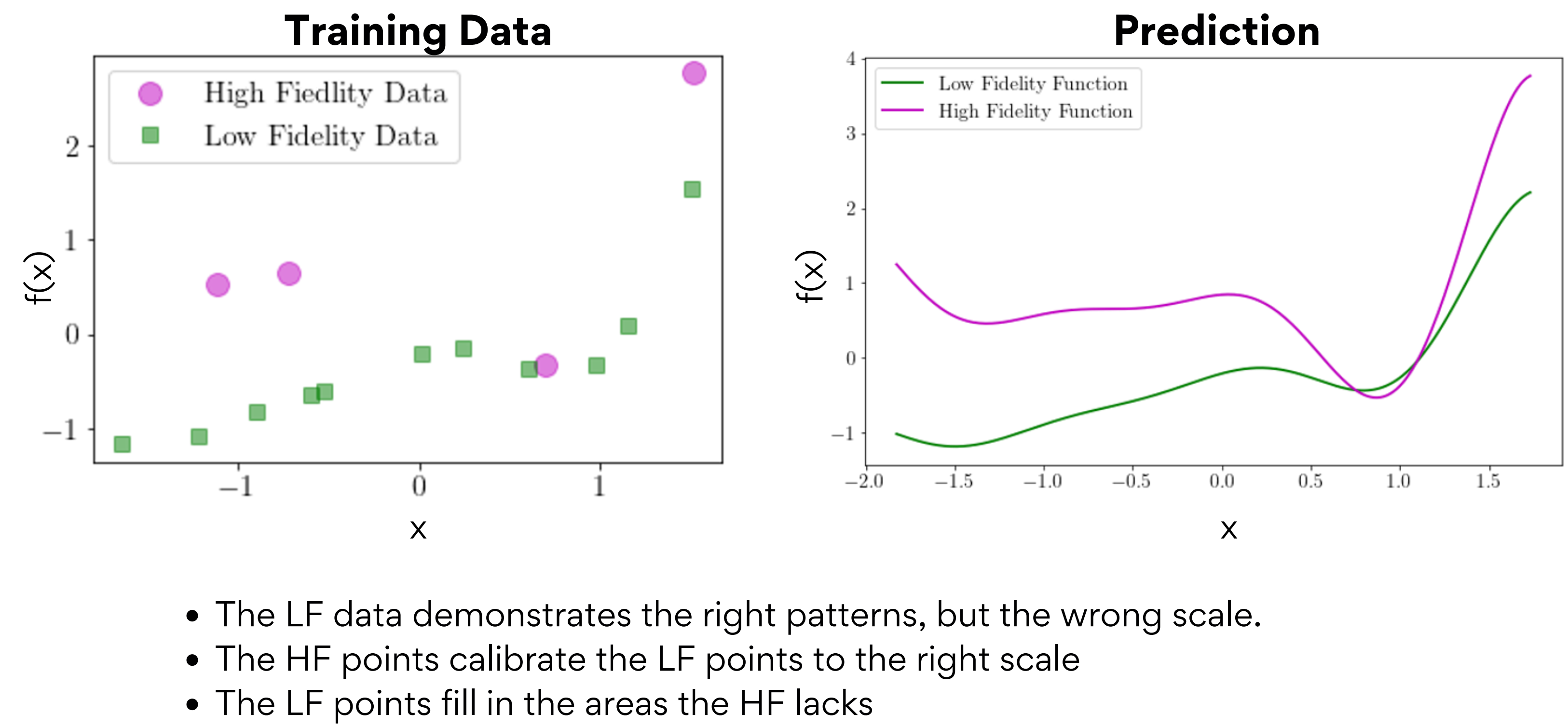


METHODS



- Finally used Linear Calibrated LF Data (653) with HF Data (12/14) for training and prediction
- "Leave One Out" method for validation

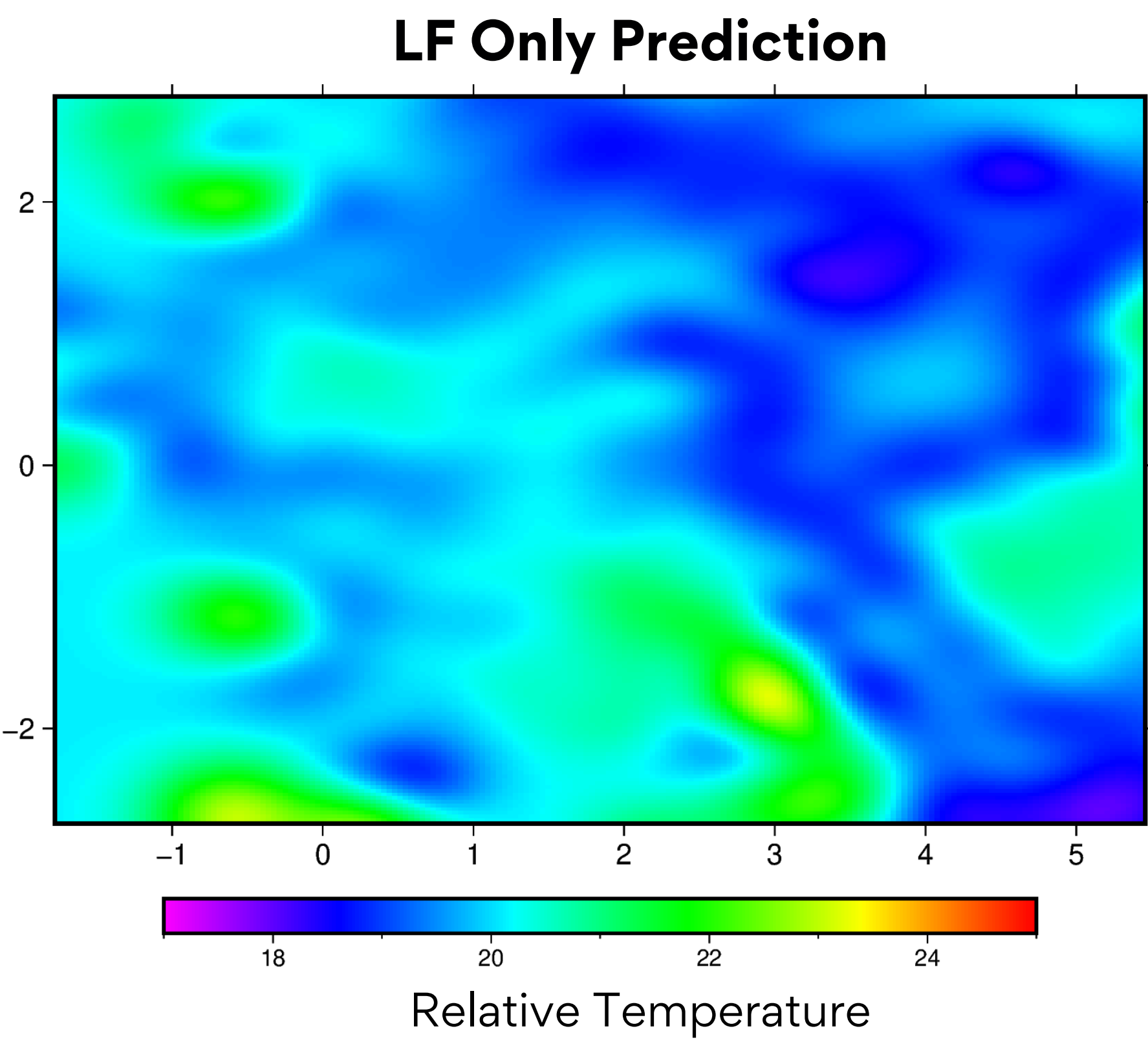
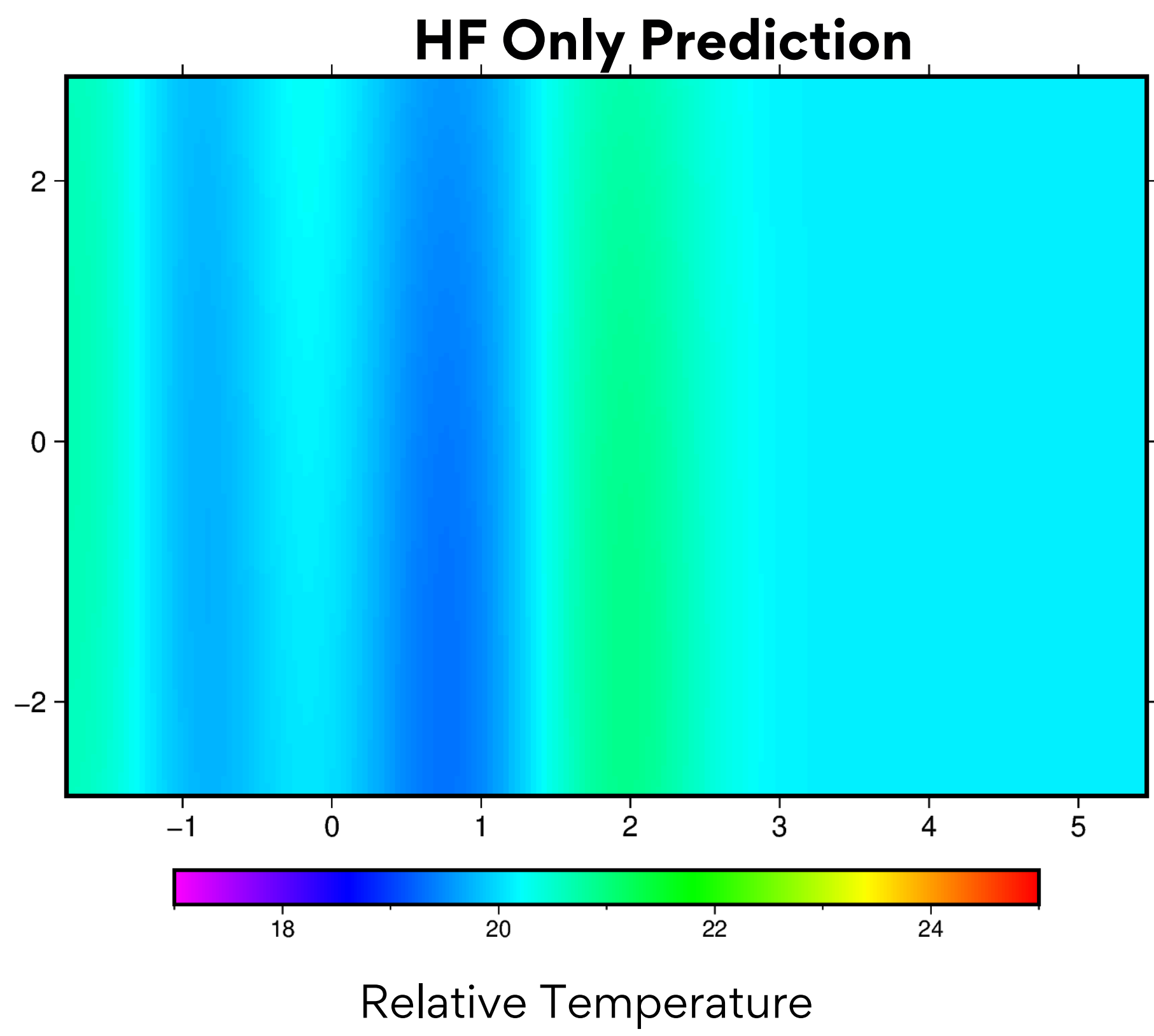
ANALOGY: WHY MULTI-FIDELITY?



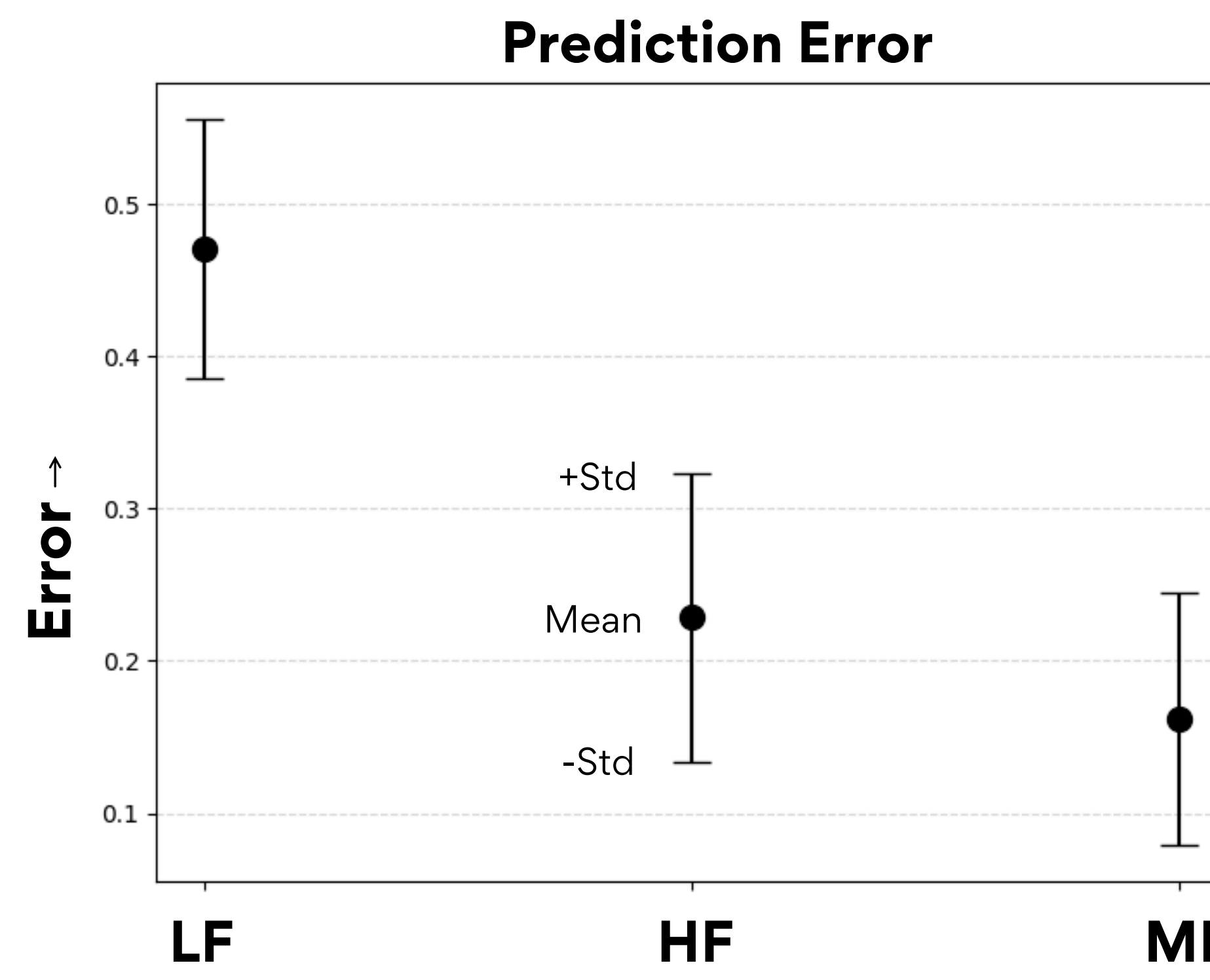
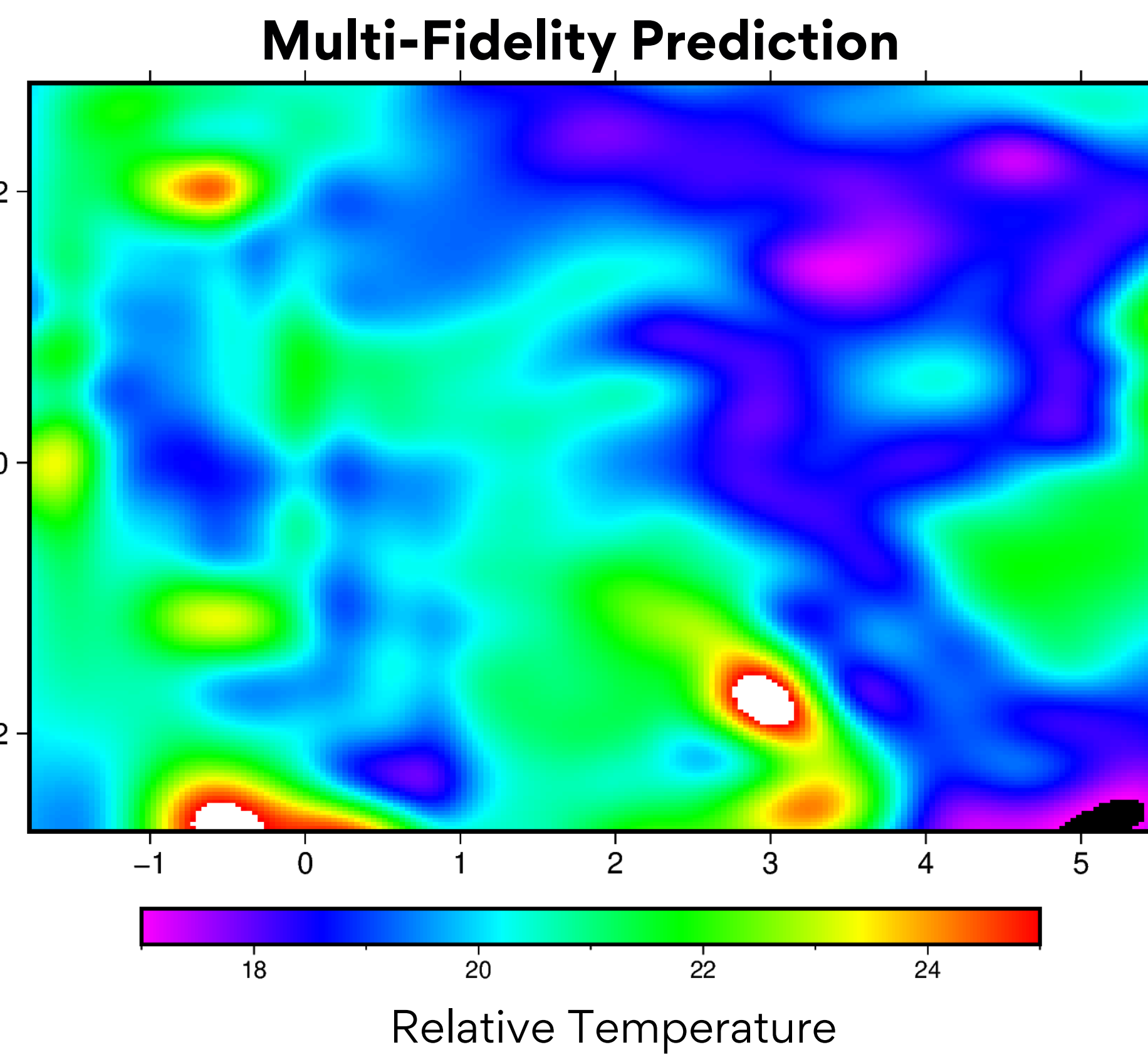
- The LF data demonstrates the right patterns, but the wrong scale.
- The HF points calibrate the LF points to the right scale
- The LF points fill in the areas the HF lacks

RESULTS

- Initial:** Prediction only using either High Fidelity or calibrated Low Fidelity data only



- Final:** Prediction only using multi-fidelity data



SUMMARY

- Multi-fidelity GPR outperformed single-fidelity models, reducing SST prediction error and providing more reliable uncertainty estimates.
- Combining satellite (LF) & buoy sensors (HF) improved spatial coverage while preserving local accuracy.

FUTURE WORK

- Incorporate Thermocline By obtaining Depth Data
- Expand high-fidelity sensor coverage
- Expand the work into a peer reviewed publication

REFERENCES

NOAA. What is a thermocline? [Internet]. Noaa.gov. 2019. Available from: <https://oceanservice.noaa.gov/facts/thermocline.html>
 Meng, X., Babaei, H., and Karniadakis, G. E., 2021, Multi-fidelity Bayesian neural networks: Algorithms and applications, Journal of Computational Physics, <https://doi.org/10.1016/j.jcp.2021.110361>
 Williams CK, Rasmussen CE. Gaussian processes for machine learning. adaptive computation and machine learning series.

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